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Notifications



1. Lecture 5

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5-1

5/5 points (graded)
Consider the system

$$\begin{aligned} \dot{x}_1 &= 6x_1 - x_2 - 4x_3 + x_4 \\ \dot{x}_2 &= 4x_1 - 6x_2 + 5x_3 - x_4 \\ \dot{x}_3 &= -6x_1 + x_2 - 5x_3 \\ \dot{x}_4 &= 4x_1 - 5x_2 + x_4. \end{aligned}$$

If

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{pmatrix} = A \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$$

is the vector equation for the given system, what is the fourth entry in the third row of the matrix **A**?

0

✓ Answer: 0

Solution:

Look carefully at the vector equation and the system. From the vector on the left-hand side, we see that the third row of **A** has to be related to $\dot{x}_3$. From the ordering of the variables in the vector on the left-hand side, we see that the fourth entry has to be the coefficient in front of x_4 in the equation for $\dot{x}_3$. From the system we see that this coefficient is equal to **0**.

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You have used 1 of 5 attempts

i Answers are displayed within the problem

5-2

5/5 points (graded)
Consider the system

$$\begin{aligned} \dot{x} &= x + 3y - z, \\ \dot{y} &= -2y + 4z, \\ \dot{z} &= 3z. \end{aligned}$$

Which of the following is the general solution of the system?
(Check all that apply.)

☐ $c_1e^t + c_2e^{-2t} + c_3e^{3t}$

☐ $c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + c_3 \begin{pmatrix} 7 \\ 8 \\ 10 \end{pmatrix}$

☒ $c_1e^t \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + c_2e^{-2t} \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + c_3e^{3t} \begin{pmatrix} 7 \\ 8 \\ 10 \end{pmatrix}$

(0)

(0)

(10)



Solution:

The system is $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ with

$$\mathbf{A} = \begin{pmatrix} 1 & 3 & -1 \\ 0 & -2 & 4 \\ 0 & 0 & 3 \end{pmatrix}.$$

Since the matrix is upper triangular, the eigenvalues are its diagonal entries **1**, **−2** and **3**. Computing $\text{NS}(\mathbf{A} - \mathbf{I})$ using back-substitution (the matrix is already in row-echelon

form) shows that the eigenspace of **1** has the vector $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ as a basis. Similarly, the

eigenspace of **−2** has $\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$ as a basis, and the eigenspace of **3** has $\begin{pmatrix} 7 \\ 8 \\ 10 \end{pmatrix}$ as a basis.

The general solution is

$$\mathbf{x} = c_1 e^t \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + c_2 e^{-2t} \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + c_3 e^{3t} \begin{pmatrix} 7 \\ 8 \\ 10 \end{pmatrix}.$$

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You have used 1 of 2 attempts

 Answers are displayed within the problem

5-3

8/8 points (graded)
For the system

$$\begin{aligned} \ddot{x} &= 5x + 7\dot{y} \\ \ddot{y} &= 2\dot{x} + 3y, \end{aligned}$$

which of the following is true?

(Suggestion: Convert it to a **first-order** system.)

☐ The set of solutions is a **1**-dimensional vector space.

☐ The set of solutions is a **2**-dimensional vector space.

☒ The set of solutions is a **4**-dimensional vector space.

☐ The set of solutions is not a vector space.



Solution:

The set of solutions is a **4**-dimensional vector space.

To convert this to a first-order system, introduce new function variables $\mathbf{u} := \dot{\mathbf{x}}$ and $\mathbf{v} := \dot{\mathbf{y}}$. Then the original system is equivalent to the first-order system

$$\dot{u} = 5x + 7v$$

$$\dot{v} = 2u + 3y.$$

This is a first-order homogeneous linear system of 4 ODEs in 4 unknown functions, so the set of solutions is a 4-dimensional vector space.

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5-4

10/10 points (graded)

Suppose that $\mathbf{X} = \begin{pmatrix} 2e^{3t} & 3e^{-2t} \\ 5e^{3t} & 7e^{-2t} \end{pmatrix}$ is a fundamental matrix for a system $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$. Let

$\mathbf{x}(t) = (x(t), y(t))$ be the solution satisfying the initial condition $\mathbf{x}(0) = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$.

What is $y(t)$?

65*exp(3*t)-63*exp(-2*t)



Answer: $65e^{3t} - 63e^{-2t}$

$65 \cdot \exp(3 \cdot t) - 63 \cdot \exp(-2 \cdot t)$

Solution:

The answer is $y(t) = 65e^{3t} - 63e^{-2t}$.

The general solution to the system has the form

$$\mathbf{x} = \mathbf{X}(t) \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = c_1 \begin{pmatrix} 2e^{3t} \\ 5e^{3t} \end{pmatrix} + c_2 \begin{pmatrix} 3e^{-2t} \\ 7e^{-2t} \end{pmatrix}.$$

Setting $t = 0$ and using the initial condition leads to

$$\begin{pmatrix} -1 \\ 2 \end{pmatrix} = c_1 \begin{pmatrix} 2 \\ 5 \end{pmatrix} + c_2 \begin{pmatrix} 3 \\ 7 \end{pmatrix},$$

which is the system

$$2c_1 + 3c_2 = -1$$

$$5c_1 + 7c_2 = 2,$$

whose solution is $c_1 = 13$, $c_2 = -9$. Thus the particular solution is

$$\mathbf{x} = 13 \begin{pmatrix} 2e^{3t} \\ 5e^{3t} \end{pmatrix} - 9 \begin{pmatrix} 3e^{-2t} \\ 7e^{-2t} \end{pmatrix}.$$

Taking the second coordinate function gives $y(t) = 65e^{3t} - 63e^{-2t}$.

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You have used 1 of 10 attempts

 Answers are displayed within the problem

5-5

5/5 points (graded)

Let $\mathbf{S} = \begin{pmatrix} 2 & 3 \\ 5 & 7 \end{pmatrix}$. Let $\mathbf{D} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. Let $\mathbf{A} = \mathbf{SDS}^{-1}$. What is the sum of the entries of $\mathbf{A}^{1000}$?

2

✔ Answer: 2

Solution:

The sum of the entries is **2**.

We have $\mathbf{A}^{1000} = \mathbf{SD}^{1000}\mathbf{S}^{-1}$. But $\mathbf{D}^2 = \mathbf{I}$, so $\mathbf{D}^{1000} = \mathbf{I}$. Thus

$$\mathbf{A}^{1000} = \mathbf{SD}^{1000}\mathbf{S}^{-1} = \mathbf{SIS}^{-1} = \mathbf{SS}^{-1} = \mathbf{I}.$$

The sum of its entries is **$1 + 0 + 0 + 1 = 2$** .

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1. Lecture 5

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